

2.3 Logical Conclusions

Lesson Introduction

 Benchmark	MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.		 Learning Targets	- I can complete an argument that leads to a logical conclusion. - I can determine if an argument leads to a logical conclusion.	
 Benchmark Coherence	Building On		Working Toward		
	N/A		N/A		
 Essential Questions	- How can I know if a conclusion is logical? - In what ways can I determine if an argument leads to a logical conclusion?				
 Lesson Narrative	In this lesson, students will develop arguments that lead to a logical conclusion. Students will begin by collaborating to connect logical arguments with the game of UNO. Then, they will complete UNO proofs to draw a logical conclusion.				
 Component Summary	2.3.1 Warm-Up (~5 min.)	Bell Work using deductive and inductive reasoning (<i>SE p. 94</i>)	2.3.3 Activity (20 min.)	Create logical arguments using UNO cards (<i>SE p. 96</i>)	
	2.3.2 Collaboration (20 min.)	Evaluate and create logical arguments using UNO cards (<i>SE p. 95</i>)	2.3.4 Wrap-Up (~5 min.)	Summary questions about 2.3.3 Activity (<i>SE p. 97</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- UNO Proof Task Cards (Optional) UNO Card Decks		Each UNO proof in 2.3.3 Activity can vary in level of difficulty. Work through each proof ahead of time to determine if scaffolding may be necessary or if a specific order might be provided. Not all proofs are designed to be completed by all students. The number of proofs needed will depend on how you divide the students.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not be familiar with the game UNO or may use alternate rules than the standard rules. - Students may not connect the reasons explicitly to each statement. - Students may miss a step in the proof process.	This lesson is designed to provide students with a connection between a common understanding of rules (i.e. through playing UNO) to completing a two-column proof through statements and accurate reasons.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Clarify, Critique, Correct 2.3.2 Collaboration question 7 provides students an opportunity to take a partial logical argument and fill in the appropriate blanks to make the argument work. Leveraging a Clarify, Critique, Correct protocol can help students work together to successfully complete the logical two-column argument.	MA.K12.MTR.1.1: Participate in Effortful Learning This lesson is centered around collaborative activities among student pairs or small groups. In 2.3.2 Collaboration, students work together to connect UNO steps to constructing a logical argument. In 2.3.3 Activity, students work together to apply that learning to completing UNO proofs. Utilize these activities to cultivate a community of growth mindset learners and to foster perseverance in students.
	This lesson provides UNO proofs for students to complete. Consider scaffolding some of the proofs to add more completed steps or remove previously completed steps to increase or decrease the cognitive demand of the task itself based on student need.	
Lesson Notes:		

2.3.1 Warm-Up: Bell Work

Component Overview: Students complete a Bell Work activity over the prior lesson: deductive reasoning, inductive reasoning, and counterexample.



2.3.1 Warm-Up: Bell Work

- Students at Cypress Bay High School must have a 2.0 grade point average in order to participate in sports. Oliver has a 2.0 grade point average. Oliver can play lacrosse for the Cypress Bay Lighting.

Determine whether the scenario uses deductive or inductive reasoning.

Deductive reasoning.

- Write a counterexample for the given statement.

If $\angle CRK$ is an acute angle, then $m\angle CRK = 84^\circ$.

Answers will vary.

Any statement of $m\angle CRK$ set equal to any value between 0° and 90° that is not 84° is correct.

Example: $m\angle CRK = 52^\circ$



Use student understanding to inform your instruction and to determine where during this and further lessons small group instruction could be utilized. Make a list of misconceptions and determine which should be addressed whole group and which should be addressed in a small group with those who are struggling to be proficient.

2.3.2 Collaboration: Logical Arguments Using UNO

Component Overview: Students will collaborate to connect the concept of logical arguments to the game of UNO. For this activity, students will work together in partner pairs or in small groups.



2.3.2 Collaboration: Logical Arguments Using UNO

In the game of UNO there are 112 cards. There are four sets of color cards: blue, green, red, and yellow, with digits 0 to 9. Each color has three types of specialty cards: Skip, Reverse, and Draw 2. There are two types of Wild cards: Wild and Wild Draw 4.

For each player's turn, the player matches the top card on the discard pile either by number, color, or word. If the player doesn't have anything that matches, they must pick a card from the draw pile. If the player draws a card that can be played, then the player plays that card.

1. Work with your partner to determine if the series of steps is possible in the game. Each statement represents a card played.

Yes, this series of steps is possible.

Statement	Reason
1. Green 8	1. Given
2. Blue 8	2. Matches number
3. Blue 3	3. Matches color
4. Wild	4. Changes color to yellow
5. Yellow Skip	5. Matches color

2. Add three additional statements and provide the reason.

Statement	Reason
6. <i>Yellow 2</i> <i>Answers will vary.</i>	6. <i>Matches color</i> <i>Answers will vary.</i>
7. <i>Yellow 5</i> <i>Answers will vary.</i>	7. <i>Matches color</i> <i>Answers will vary.</i>
8. <i>Red 5</i> <i>Answers will vary.</i>	8. <i>Matches number</i> <i>Answers will vary.</i>



Prior to releasing students to work on their own with their partner or group, review the narrative language that is provided prior to question 1. This will help set the stage for the activity.

Be mindful that some students may not have played the game UNO before or may play by a different set of rules than the standard UNO rules. Consider surveying students to determine what they know and do not know about the game. Show a basic review of the game if necessary.

This collaboration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. The idea of proofs is often intimidating for students. This lesson is meant to develop students' understanding of a logical argument and how one step in a proof leads to the next (in most cases). Allow students to struggle and to revise their thinking throughout the collaboration. Observe and listen as students work with their partners while providing a safe environment for the students to be wrong and to ask questions. Additional scaffolding questions are provided, if needed, but this should be primarily student-directed.



Why do you think it is possible for there to be more than one correct answer for the next step? Why are "Yellow 2" and "Blue Skip" both correct answers? Would both correct answers have the same reason? Why or why not?

2.3.2 Collaboration: Logical Arguments Using UNO (continued)

3. Ask your partner to verify that your steps are valid.
4. The series of steps shown is not possible. Correct the error.

Answers will vary.

Statement	Reason
1. Red 4	1. Given
2. Red 9	2. Matches color
3. Yellow 9	3. Matches number
4. Green 4 9	4. Changes color to green <i>Matches Number</i>
5. Red 4 9	5. Matches number
6. Red Reverse	6. Matches color

5. Ensure that you and your partner found the same error and that the corrections are both correct.

Yolanda made the following corrections.

Statement	Reason
1. Red 4	1. Given
2. Red 9	2. Matches color
3. Yellow 9	3. Matches number
4. Green 4 Blue 9	4. Changes color to green Matches number
5. Red 4 Blue Skip	5. Matches number color
6. Red Reverse Blue 5	6. Matches color



If available, consider using physical UNO cards as part of this activity. This can help kinesthetic learners be able to engage in a hands-on experience while allowing visual learners to see the statement in action and connect the correct reason from the card game to the table.



For an additional challenge, have students create another series of steps that are not possible for their partner to find the error and make corrections.

2.3.2 Collaboration: Logical Arguments Using UNO (continued)

6. Explain why you or your partner's corrections may not match Yolanda's corrections.

Answers will vary. There are many options for how this could be corrected. My partner changed the Green 4 to a Yellow 3 and had a completely different set of reasons as well.

7. Work with your partner to create a logical argument that leads to a Red Draw 2.

Answers will vary.

Statement	Reason
1. Green 5	1. Given
2. Yellow 5	2. Matches number
3. Yellow 4	3. Matches color
4. Green 4	4. Matches number
5. Green 9	5. Matches color
6. Green 8	6. Matches color
7. Green 3	7. Matches color
8. Red 3	8. Matches number
9. Red 8	9. Matches color
10. Red Draw 2	10. Matches color

8. With your partner, use the template provided by your teacher to fill in your reasons for 5, 8, and 10. Then switch your paper with another pair. Work with your partner to fill in the missing statements and reasons on the other pair's paper.

Answers will vary.

9. Work as a group of four to make sure that the two completed templates correctly lead to the conclusion.



As you circulate consider asking students to share their answers to question 6. This question can vary based on how each pair chose to fix the original errors. Include in the discussion how it is possible for there to be multiple correct ways to fix the error and the importance of having an accurate reason that supports the statement. Use the opportunity to listen and build student confidence.



Consider using the digital template to provide scaffolds for students who may need support or to increase the challenge of this activity for students who are ready. This can be done by removing statements or changing the color provided, or only providing the given and the last step. Whichever you choose, work through it again to ensure that the answer will work with the number of statements provided in the template.

2.3.3 Activity: Taking Turns

Component Overview: Students will work together to complete a series of proofs using UNO cards. For this activity, students should work in partner pairs or in small groups. The templates for the cards are located in the teacher guide.



2.3.3 Activity: Taking Turns

Complete the UNO proofs. All UNO cards are available except for the Wild or Wild Draw 4 card. The only time a Wild card can be used is when it is already written in the proof.

Each proof must have at least three number changes and three color changes.

Answers will vary.



For this activity, it is suggested that students work with partners and that each pair starts a proof. Then, the proof is passed to the next pair, etc. The goal is for all proofs to be completed by the class, not that each student pair completes each proof. Use the activity to cultivate a community of growth mindset learners and to foster perseverance by reminding the class that the goal is to work together to complete as many proofs as possible. For classes where students may struggle consider lowering the number of proofs and changing from partners to groups of three or four. For Honors, utilize all of the proofs or create some on your own that have an increased difficulty.

Alternative Methods:

- Post the templates up around the room and have students complete one, two, or three steps before moving to the next.
- Have each pair of students complete a proof in its entirety and then share with the class.

2.3.4 Wrap-Up: Reflection

Component Overview: In this activity, students will answer three questions to summarize the activity. These questions can be done independently or with their partner or group from the prior activity.



2.3.4 Wrap-Up: Reflection

1. How many of the UNO proofs did the class correctly complete?
Answers will vary.
2. Which steps in the UNO proof were the easiest to complete?
Answers will vary.
3. Which steps in the UNO proof were hardest to complete?
Answers will vary.



The information provided from the student here, coupled with what was observed during instruction, can provide information for how engaged each student was during the lesson. This provides an individual accountability of a student's engagement (MA.K12.MTR.1.1). Consider reviewing student responses to consider which students may need further support in effortful learning. Also consider writing an encouraging note to students about their engagement.